

Tirtho Mondal

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Career Objective

My goal is to contribute to higher education through effective teaching, student mentorship, and continuous learning in the field of Computer Science and Engineering. I aim to help students build strong academic and practical skills for successful careers while also contributing to research, innovation, and academic development.

Education

Khulna University of Engineering & Technology (KUET) B.Sc. in Computer Science and Engineering	2022 – 2026 CGPA: 3.72 / 4.00
Kapilmuni College Higher Secondary Certificate (H.S.C.) in Science	2018 – 2020 GPA: 5.00 / 5.00
Kapilmuni Sahachari Vidya Mandir Secondary School Certificate (S.S.C.) in Science	2016 – 2018 GPA: 5.00 / 5.00

Research Interests

Optimization Theory | Game Theory | Computer Vision | Natural Language Processing | Machine Learning | Artificial Intelligence

Publications

- [1] D. Das, A. D. Das, F. Sadaf, A. Uddin, and **T. Mondal**, “Real-Time Assistive Navigation for the Visually Impaired: A Scalable Approach for Indoor and Outdoor Mobility,” *arXiv preprint arXiv:2504.20976*, 2025 [🔗](#)

Projects

3D Cafeteria Simulation 🎮	C++ OpenGL 3.3 GLFW GLM
Developed an interactive 3D cafeteria environment utilizing texture mapping, Phong/Gouraud shading, Bézier curves, and animated characters driven by sine-wave walking motions.	
Smart Navigation for the Visually Impaired 🎮	Python Flutter
Built an assistive navigation system utilizing computer vision and sensor fusion for indoor/outdoor mobility, integrating real-time audio and haptic feedback mechanisms.	
TutorFinder 🎮	Laravel PHP MySQL
Designed a web platform that connects students with local tutors through comprehensive profile-based matching algorithms and integrated scheduling features.	
HouseHold Services 🎮	Android Java
Created an Android application focused on local service delivery, allowing users to seamlessly book, track, and manage household service requests in real time.	
CSE Family 🎮	JavaFX
Formed a desktop application tailored for the student community to centralize academic resource sharing, peer-to-peer discussions, and department updates.	
MobiCash 🎮	C++
Implemented a mobile banking simulation handling core financial workflows, including secure transactions, mobile recharges, and user account management.	
Laundry Shop Management 🎮	SQL
Engineered a relational database system designed to systematically manage customer profiles, order workflows, payment processing, and overall shop operations.	

Technical Skills

Languages	C, C++, Python, JavaScript, PHP, SQL, HTML/CSS, Java, Dart
Frameworks	Laravel, Android SDK, Flutter
Tools	Git/GitHub, VS Code, IntelliJ IDEA, Android Studio, Jupyter Notebook, L ^A T _E X
Research	NumPy, Pandas, OpenCV, Sensor Fusion
Concepts	Machine Learning, Deep Learning, Computer Vision, NLP, REST APIs, OOP, Design Patterns
Productivity	MS Word, Excel, PowerPoint

Honours & Awards

- **Dean’s List:** Academic Years 2021–2022 and 2022–2023, Department of CSE, KUET

Languages

Bengali: Native | **English:** Fluent